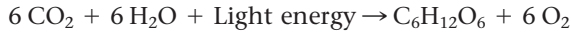


and H₂O. We can summarize the complex series of chemical reactions in photosynthesis with this chemical equation:

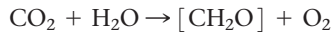


We use glucose (C₆H₁₂O₆) here to simplify the relationship between photosynthesis and respiration, but the direct product of photosynthesis is actually a three-carbon sugar that can be used to make glucose. Water appears on both sides of the equation because 12 molecules are consumed and 6 molecules are newly formed during photosynthesis. We can simplify the equation by indicating only the net consumption of water:



Writing the equation in this form, we can see that the overall chemical change during photosynthesis is the reverse of the one that occurs during cellular respiration (see Concept 9.1). It is important to realize that *both* of these metabolic processes occur in plant cells. However, as you will soon learn, chloroplasts do not synthesize sugars by simply reversing the steps of respiration.

Now let's divide the photosynthetic equation by 6 to put it in its simplest possible form:



Here, the brackets indicate that CH₂O is not an actual sugar but represents the general formula for a carbohydrate (see Concept 5.2). In other words, we are imagining the synthesis of a sugar molecule one carbon at a time (with six repetitions theoretically adding up to a glucose molecule: C₆H₁₂O₆). Let's now see how researchers tracked the elements C, H, and O from the reactants of photosynthesis to the products.

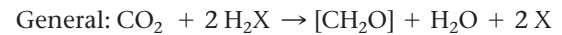
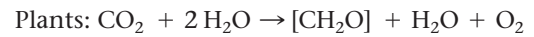
The Splitting of Water: Scientific Inquiry

One of the first clues to the mechanism of photosynthesis came from the discovery that the O₂ given off by plants is derived from H₂O and not from CO₂. The chloroplast splits water into hydrogen and oxygen atoms. Before this discovery, the prevailing hypothesis was that photosynthesis split carbon dioxide (CO₂ → C + O₂) and then added water to the carbon (C + H₂O → [CH₂O]). That hypothesis predicted that the O₂ released during photosynthesis came from CO₂, an idea challenged in the 1930s by C. B. van Niel, of Stanford University. Van Niel was investigating photosynthesis in bacteria that make their carbohydrate from CO₂ but do not release O₂. He concluded that, at least in these bacteria, CO₂ is not split into carbon and oxygen. One group of bacteria used hydrogen sulfide (H₂S) rather than water for photosynthesis, forming yellow globules of sulfur as a waste product (these globules are visible in Figure 10.2e). Here is the chemical equation for photosynthesis in these sulfur bacteria:



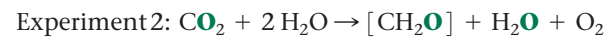
Van Niel reasoned that the bacteria split H₂S and used the hydrogen atoms to make sugar. He then generalized that idea,

proposing that all photosynthetic organisms require a hydrogen source but that the source varies:



Thus, van Niel hypothesized that plants split H₂O as a source of electrons from hydrogen atoms, releasing O₂ as a by-product.

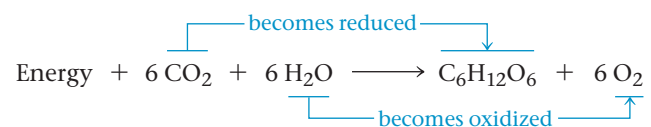
Nearly 20 years later, scientists confirmed van Niel's hypothesis by using oxygen-18 (¹⁸O), a heavy isotope, as a tracer to follow the path of oxygen atoms during photosynthesis. The experiments showed that the O₂ produced by plants was labeled with ¹⁸O *only* if water was the source of the tracer (experiment 1). If the ¹⁸O was introduced to the plant in the form of CO₂, the label did not turn up in the released O₂ (experiment 2). In the following summary, green denotes labeled atoms of oxygen (¹⁸O):



A significant result of the shuffling of atoms during photosynthesis is the extraction of hydrogen from water and its incorporation into sugar. The waste product of photosynthesis, O₂, is released to the atmosphere. **Figure 10.4** shows the fates of all atoms in photosynthesis.

Photosynthesis as a Redox Process

Let's briefly compare photosynthesis with cellular respiration. Both processes involve redox reactions. During cellular respiration, energy is released from sugar when electrons associated with hydrogen are transported by carriers to oxygen, forming water as a by-product (see Concept 9.1). The electrons lose potential energy as they “fall” down the electron transport chain toward electronegative oxygen, and the mitochondrion harnesses that energy to synthesize ATP (see Figure 9.14). Photosynthesis reverses the direction of electron flow. Water is split, and its electrons are transferred along with hydrogen ions (H⁺) from the water to carbon dioxide, reducing it to sugar.



▼ **Figure 10.4 Tracking atoms through photosynthesis.** The atoms from CO₂ are shown in magenta, and the atoms from H₂O are shown in blue.

